University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Yr. (Even Sem.), Term II Examination, 2016-17

Subject: English for Professional Purposes-II Subject Code: AHE 4083
Time: 50 Minutes Max. Marks: 15

Notes:

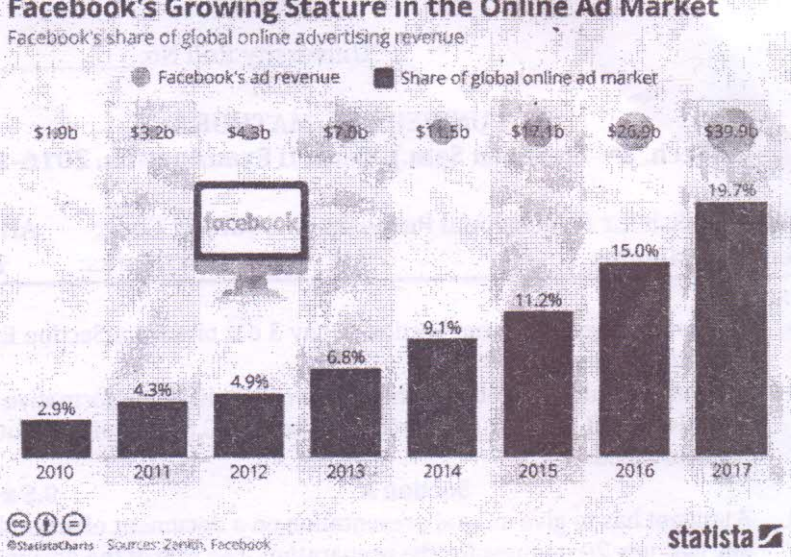
- Answer all questions from Section A, any 3 out of 4 from Section B; Section C has internal choices.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 60 words) and Section C (150 words) should strictly be maintained.

Section A 0.5 x 6=3

1. A student has to give an oral presentation on a document of around 25 pages. She has only 20 minutes for the preparation. The title of the document is *Case Studies in Intensive Reading Skills*. Which Skill of Reading is most apt for this assignment?
2. Rewrite the following sentences with brevity: (you may add or delete the words but you cannot change the meaning)
   "The young man had no idea about what was going to happen with him the next day."
3. Give any one reason to support the argument that 'good listening power is essential for a good speaker'.
4. A presenter touches his forehead several times during a power point presentation. Can this be an example of poor non-verbal mannerism? Give one sentence argument to support your answer.
5. Give any two situations where one cannot do without good negotiation skills.
6. In a cricket match, the Required Run Rate for 'Team A' increases after every over till over number 25, then in decreases till over number 50. Make a graph to show this. Indicate the Over on X-axis and increasing Required Run Rate on Y-axis.

Section B 2x3=6

1. Make the first two slides (excluding the title slide) on the topic, *Placements at GLA University*.
2. Write any four analytical points/observations on the chart entitled *Facebook's Growing Stature in the Online Ad Market.* (see. Picture)
3. You are a senior correspondent of The Hindu. You have taken an interview of the governor of the Reserve Bank of India. Write the excerpt of the interview that may be an example of skill of assertiveness.



4. Write any four ways of improving our listening skills.

Section C **3x2=6**

1. Combine the following sentences into one, following the principles of brevity:
   a. Sometimes Krishan went running with Harsha. He was a good athlete. He was on the track team at school.
   b. Astha brought some candy back from Europe. It wasn't shaped like American candy. The candy tasted kind of strange to her.
   c. Government leaders like to mention the creation of new jobs. They claim that these new jobs indicate a strong economy. They don't mention that low-wage jobs without benefits and security have replaced many good jobs.

Or

Give the lay out/format of a business letter.

2. How does the story of *Animal Farm*demonstrate the failure of institutional politics?

Or

What is the role of the sea in *The Old Man and the Sea*?

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Set A University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017-18

Subject: English for Professional Purposes–II Subject Code: AHE 4083

Time: 45 Minutes Max. Marks: 40

VERBAL APTITUDE

Choose the most appropriate option for Q. No. 1-6.

1. There appeared/seemed to be some difficulty in fixing a date for the meeting. Which one is the adverb?
   A. There B. Appeared C. difficulty
   D. date E. none of the above
2. An adverbial can modify a/an
   A. verb B. Adjective C. adverb
   D. clause E. all of the above
3. The children really liked the museum, _____ the dinosaurs.
   A. specially B. Especially C. while
   D. none but E. none of the above
4. _____ such an honest teacher.
   A. Nowhere I have encountered
   B. Nowhere have I ever encountered
   C. Nowhere have I encounter
   D. Nowhere I have had encountered
   E. none of the above
5. _____ started mowing the lawn _____ it started raining.
   A. As soon as he had / that B. No sooner had he / than
   C. No sooner he had / than D. Although he / but
   E. none of the above
6. The stillness of the temple was (awfully, awful). The temple was (awfully, awful) still.
   A. awful/awfully B. awfully/awful
   C. awful/awful D. awfully/awfully
   E. none of the above

Decide which sentence is the most preferable in respect to grammar/ meaning and usage, suitable for formal writing.

7. A. The general manager was really angry when one of his employees blurted in the secret.
   B. The general manager was really angry when one of his employees blurted on the secret.
   C. The general manager was really angry when one of his employees blurted away the secret.
   D. The general manager was really angry when one of his employees blurted out the secret.
   E. The general manager was really angry when one of his employees blurted up the secret.
8. A. Harish told us about some fight on a bus in which his friends had duffed on the bus conductor.
   B. Harish told us about some fight on a bus in which his friends had duffed in the bus conductor.
   C. Harish told us about some fight on a bus in which his friends had duffed up the bus conductor.
   D. Harish told us about some fight on a bus in which his friends had duffed away the bus conductor.
   E. Harish told us about some fight on a bus in which his friends had duffed under the bus conductor.
9. A. The arrival of the charity van set off a minor riot as villagers scrambled for a share of the aid.
   B. B. The arrival of the charity van set in a minor riot as villagers scrambled for a share of the aid.
   C. The arrival of the charity van set out a minor riot as villagers scrambled for a share of the aid.
   D. The arrival of the charity van set away a minor riot as villagers scrambled for a share of the aid.
   E. The arrival of the charity van set on a minor riot as villagers scrambled for a share of the aid.
10. A. She'd gladly have gone anywhere to get out from the monotonous life.
    B. She'd gladly have gone anywhere to get in from the monotonous life.
    C. She'd gladly have gone anywhere to get away from the monotonous life.
    D. She'd gladly have gone anywhere to get off from the monotonous life.
    E. She'd gladly have gone anywhere to get into from the monotonous life.
11. A. The high divorce figures don't seem to be putting people out marriage.
    B. The high divorce figures don't seem to be putting people on marriage.
    C. The high divorce figures don't seem to be putting people in marriage.
    D. The high divorce figures don't seem to be putting people of marriage.
    E. The high divorce figures don't seem to be putting people off marriage.

Choose the correct option.

12. The right to self government
    A. Autobiography B. Autocracy C. Autonomy
    D. Bureaucracy E. Aristocracy
13. One that belong to the same time period
    A. Contemplate B. Compatriot C. Contemporary
    D. Contentious E. None of the above
14. One who speaks two languages
    A. Monolingual B. Polyglot C. Linguist
    D. Orator E. None of the above
15. One who easily believes what others say
    A. Incredible B. Creditable C. Credentials
    D. Credulous E. None of the above
16. One who eats human flesh
    A. Hannibal B. Carnivorous C. Cannabis
    D. Cannibal E. Carnival

Pick out the most effective words to make the sentence complete.

17. You __ when I ___ the television yesterday at eight p.m.
    A. were telephoning, watched
    B. was telephoning, was watching
    C. had telephoned, was watched
    D. telephoned, was watching
    E. telephones, watched
18. He _____ in New York _____ he left the university.
    A. is living, since B. has been living, for
    C. has been living, since D. lives, for
    E. is living, for

Read each sentence to find out whether there is any grammatical or idiomatic error in it.

19. The police finally (A)/ catched up with (B)/ the notorious robber who (C)/ escaped from jail. (D)/ No error (E)
20. The gold necklace (A)/ looked very beautiful on (B)/ her and complemented (C)/ her dress too. (D)/ No error (E)
21. Indeed, it is unlikely that any kind (A)/ of censorship would be particularly successful, as the problem lies (B)/ as much with underlying attitude(C)/ as it does with the adverts or jokes themselves (D)./ No error (E)
22. People who is familiar (A)/ with Aran patterns will recognize (B)/ this technique on the sweater (C)/ Aran stitching is an old art form (D)/ No error (E)

Fill in the blanks with the most suitable option:

23. If they _____ now, they _____ in New Delhi by lunchtime.
    A. leave/will B. leave/will be C. left/would
    D. have left/would have E. will leave/will be
24. If he _____ more careful, he _____ caught.
    A. had been/ wouldn't have B. had / wouldn't have
    C. has been/will D. was/wouldn't
    E. None of these
25. Your pet _____ so fat if you _____ feed it so often.
    A. will not/will not B. would not/was not
    C. would not be/ did not D. would not been/did not
    E. None of these
26. If you _____ the money ____ lent it to me?
    A. had/ you would have B. had been/you would have
    C. had had/you would have D. had had/would you have
    E. have/you will have

Choose the correct option.

27. Deepak is negligent _____ attending his classes
    A. in B. to C. about
    D. towards E. none of these
28. He stands_____ what he has done.
    A. at B. for C. upon
    D. on E. none of these
29. I hate sitting _____ as he always smells of garlic.
    A. beside B. besides C. along
    D. at E. none of these
30. Savita kept talking loudly oblivious___ the fact that it was the college library and not the students' common room.
    A. for B. to C. about
    D. of E. none of these
31. Which idiom means "to talk in an indirect manner?"
    A. To get off on the wrong foot
    B. To be off the hook C. Once in a blue moon
    D. Beating around the Bush E. Cut to the Chase
32. The idiom "to cross one's fingers" means:
    A. To work hard B. Unlikely to succeed
    C. Waiting for the result anxiously
    D. To give a hint E. None of the above

Read the passage carefully and answer the questions that follow.

The happy man is the man who lives objectively, who has free affections and wide interests, who secures his happiness through these interests and affections and through the fact that they in turn make him an object of interest and affection to many others. To be the recipient of affection is a potent cause of happiness, but the man who demands affection is not the man upon whom it is bestowed. The man who receives affection is, speaking broadly, the man who gives it. But it is useless to attempt to give it as a calculation, in the way in which one might lend money at interest, for a calculated affection is not genuine and is not felt to be so by the recipient. What then can a man do who is unhappy because he is encased in self? So long as he continues to think about the causes of his unhappiness, he continues to be self-centered and therefore does not get outside it. It must be by genuine interest, not by simulated interests adopted merely as a medicine. Although this difficulty is real, there is nevertheless much that he can do if he has rightly diagnosed his trouble. If for example, his trouble is due to a sense of sin, conscious or unconscious, he can first persuade his conscious mind that he has no reason to feel sinful, and then proceed, to plant this rational conviction in his unconscious mind, concerning himself meanwhile with some more or less neutral activity. If he succeeds in dispelling the sense of sin, it is possible that genuine objective interests will arise spontaneously. If his trouble is self-pity, he can deal with it in the same manner after first persuading himself that there is nothing extraordinarily unfortunate in his circumstances. If fear is his trouble, let him practice exercises designed to give courage. Courage has been recognized from time immemorial as an important virtue, and a great part of the training of boys and young men has been devoted to producing a type of character capable of fearlessness in battle. But moral courage and intellectual courage have been much less studied. They also, however, have their technique. Admit to yourself every day at least one painful truth, you will find it quite useful. Teach yourself to feel that life would still be worth living even if you were not, as of course you are, immeasurably superior to all your friends in virtue and in intelligence. Exercises of this sort prolonged through several years will at last enable doing, free you from the empire of fear over a very large field.

33. According to the passage, calculated affection
    A. appears to be false and fabricated
    B. makes other person to love you
    C. turns into permanent affection over a period of time
    D. leads to self-pity
    E. gives a feeling of courage
34. Who according to the passage is the happy man?
    A. Who is encased in self
    B. Who has free affection and wide interests
    C. Who is free from worldly passions
    D. Who has externally centred passions
    E. None of these
35. Which of the following statements is NOT TRUE in the context of the passage?
    A. The happy man has wide interests.
    B. Courage has been recognized as an important virtue.
    C. Unhappy man is encased in self.
    D. A man who suffers from the sense of sin must tell himself that he has no reason to be sinful.
    E. Issue of intellectual courage has been extensively studied.
36. Which of the following virtues, according to the passage, has been recognized for long as an important virtue?
    A. Patriotism B. Sacrifice C. Courage
    D. Self-consciousness E. None of these
37. Which of the following words is SIMILAR in meaning of the word 'bestowed' as used in the passage?
    A. Conferred B. Accommodated C. Trusted
    D. Withdrawn E. Directed
38. Which of the following, according to the passage, has not been studied much?
    A. Feeling of guilt and self-pity
    B. The state of mind of an unhappy man
    C. How to get absorbed in other interests
    D. Moral and intellectual courage
    E. None of these
39. What should a man do who is suffering from the feeling of self-pity?
    A. He should control his passions and emotions.
    B. He should persuade himself that everything is alright in his circumstances.
    C. He should seek affection from others.
    D. He should develop a feeling of fearlessness.
    E. He should consult an expert to diagnose his trouble.
40. What happens to a man who demands affection?
    A. His feelings are reciprocated by others.
    B. He tends to take a calculated risk.
    C. He becomes a victim of a vicious circle.
    D. He takes affection for granted from others.
    E. None of these

Set B

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017-18

Subject: English for Professional Purposes–II | Subject Code: AHE 4083

Time: 45 Minutes | Max. Marks: 40

VERBAL APTITUDE

Choose the most appropriate option for Q. No. 1-6.

1. There's still time for us to get to the cinema before the film starts. Which one is the adverb?
   A. still B. to get C. before
   D. time E. none of the above
2. _____ such an honest teacher.
   A. Nowhere I have encountered
   B. Nowhere have I ever encountered
   C. Nowhere have I encounter
   D. Nowhere I have had encountered
   E. none of the above
3. He uses a/an_____ adapted wheelchair.
   A. specially B. Especially C. most
   D. nothing E. none of the above
4. ____ those words ____ .all the newspapers published them.
   A. As soon as he had uttered/ that
   B. No sooner had he uttered/ than
   C. No sooner he had uttered/ than
   D. Although he uttered/ but
   E. none of the above
5. An adverbial can modify a/an
   A. verb B. Adjective C. adverb
   D. clause E. all of the above
6. The programmer's techniques were _____. The manager said that his work was_____ done.
   A. well/good B. well/well C. good/good
   D. good/well E. none of the above

Decide which sentence is the most preferable in respect to grammar/ meaning and usage, suitable for formal writing.

7. A. The Chief Minister initiated a campaign to beef up security.
   B. The Chief Minister initiated a campaign to beef upon security.
   C. The Chief Minister initiated a campaign to beef on security.
   D. The Chief Minister initiated a campaign to beef out security.
   E. The Chief Minister initiated a campaign to beef in security.
8. A. My friend came out with chicken pox this weekend.
   B. My friend came in with chicken pox this weekend.
   C. My friend came away with chicken pox this weekend.
   D. My friend came down with chicken pox this weekend.
   E. My friend came in with chicken pox this weekend.
9. A. In India a house is broken in every 24 seconds.
   B. In India a house is broken to every 24 seconds.
   C. In India a house is broken into every 24 seconds.
   D. In India a house is broken on every 24 seconds.
   E. In India a house is broken up every 24 seconds.
10. A. She could not bear the music and turned up the volume.
    B. She could not bear the music and turned down the volume.
    C. She could not bear the music and turned off the volume.
    D. She could not bear the music and turned in the volume.
    E. She could not bear the music and turned on the volume.
11. A. We ended in renting a movie instead of going to the theatre.
    B. We ended upon renting a movie instead of going to the theatre.
    C. We ended on renting a movie instead of going to the theatre.
    D. We ended up renting a movie instead of going to the theatre.
    E. We ended away renting a movie instead of going to the theatre.

Choose the correct option.

12. The art of beautiful writing.
    A. Cartography B. Lithography C. Calligraphy
    D. All of the above E. None of the above
13. Something that which can be heard.
    A. Audible B. Cordial C. Edible
    D. Culpable E. None of the above
14. One who is completely self satisfied.
    A. Competent B. Compliance C. Complacent
    D. Arrogance E. None of the above
15. A statement that is open to more than one interpretation.
    A. Ambivalent B. Amphibious C. Ambiguous
    D. Antigen E. None of the above
16. Murder of a brother.
    A. Fungicide B. Pesticide C. Suicide
    D. Fratricide E. None of the above

Pick out the most effective words to make the sentence complete.

17. I _____ here _____ twenty years next month.
    A. working, since B. will be working, since
    C. will have been working, since
    D. will have been working, for
    E. will be working, for
18. I _____ with my sister _____ I find an apartment.
    A. am living, until B. have been living, since
    C. had lived, since D. had been living, since
    E. am living, since

Read each sentence to find out whether there is any grammatical or idiomatic error in it.

19. The teacher told me (A)/ to give a thoroughly (B)/ reading to every (C)/ chapter of the book (D).No error. (E)
20. The bank has (A) decided to write (B) of the debts in return (C) for fresh deposits (D). No error (E)
21. One of the main consequences of this is (A)/ that the Front national do not benefit from any (B)/ tactical vote transfer from other parties — a crucial (C)/ stepping stone to success in a two-round electoral system (D)/ No error (E)
22. To begin with, it will reopen (A)/ a historic route that reconnect (B)/ South Asia to Central Asia, in the way it was (C)/ before the British Empire sealed it off (D)/ No error (E)

Fill in the blanks with the most suitable option.

23. You _____ failed the exam if you _____ a bit harder.
    A. would have/would studied B. wouldn't have/studied
    C. wouldn't have/had studied
    D. would have been/had studied E. will/study
24. You _____ to the party as long as you _____ home before midnight.
    A. will go/ will come B. may went/come
    C. can go/come D. go/will come
    E. could go/had come
25. They _____ up a lie if they _____ it would help me.
    A. would have made/had thought
    B. would made/had thought
    C. would make/had thought
    D. would have been make/had thought
    E. will made/thought
26. If I _____ to hold a speech I _____ pretty nervous.
    A. had had/ would have B. had/ would have been
    C. had/ would be D. have/ would be
    E. was/would be

Choose the correct option.

27. After the Mahabharata war, the Pandavas renounced worldly life and made ____ the Himalayas.
    A. To B. for C. into
    D. towards E. none of these
28. A great writer adopts a style appropriate ____his/her subject.
    A. with B. for C. to
    D. into E. none of these
29. We are looking forward ____ your cooperation.
    A. of B. for C. about
    D. to E. none of these
30. He has aptitude _____
    A. in B. for C. about
    D. of E. none of these
31. Which idiom means "something that first seems negative but turns out to be positive in the end."
    A. A blessing in disguise B. Once in a blue moon
    C. The buck stops here D. Every cloud has a silver lining
    E. To get off on the wrong foot
32. The idiom "Once in a blue moon" means:
    A. A moon that appears blue B. Very rarely
    C. Doing something only once D. Not always
    E. None of the above

Read the passage carefully and answer the questions that follow.

Management is a set of processes that can keep a complicated system of people and technology running smoothly. The most important aspects of management include planning, budgeting, organizing, staffing, controlling, and problem-solving. Leadership is a set of processes that creates organisations in the first place or adapts them to significantly changing circumstances. Leadership defines what the future should look like, aligns people with that vision, and inspires them to make it happen despite the obstacles. This distinction is absolutely crucial for our purposes here: Successful transformation is 70 to 90 per cent leadership and only 10 to 30 per cent management. Yet for historical reasons, many organisations today don't have much leadership. And almost everyone thinks about the problems here as one of managing change. For most of this century, as we created thousands and thousands of large organizations for the first time in human history, we didn't have enough good managers to keep all those beaurocracies functioning. So many companies and universities developed management programmes, and hundreds and thousands of people were encouraged to learn management on the job. And they did. But, people were taught little about leadership. To some degree, management was emphasized because it's easier to teach than leadership. But even more so, management was the main item on the twentieth-century agenda because that's what was needed. For every entrepreneur or business builder who was a leader, we needed hundreds of managers to run their ever growing enterprises. Unfortunately for us today, this emphasis on management has often been institutionalized in corporate cultures that discourage employees from learning how to lead. Ironically, past success is usually the key ingredient in producing this outcome.

The syndrome, as I have observed it on many occasions, goes like this: success creates some degree of market dominance, which in turn produces much growth. After a while keeping the ever larger organizations under control becomes the primary challenge. So attention turns inward, and managerial competencies are nurtured. With a strong emphasis on management but not on leadership, bureaucracy and an inward focus take over. But with continued success, the result mostly of market dominance, the problem often goes unaddressed and an unhealthy arrogance begins to evolve. All of these characteristics then make any transformation effort much more difficult. Arrogant managers can over-evaluate their current performance and competitive position, listen poorly, and learn slowly. Inwardly focused employees can have difficulty seeing the very forces that present threats and opportunities. Bureaucratic cultures an smother those who want to respond to shifting conditions. And the lack of leadership leaves no fore inside these organizations to break out of the morass.

33. Why, according to the author, is a distinction between management and leadership crucial?
    A. Leaders are reactive whereas managers are proactive.
    B. Organizations are facing problems of not getting good managers.
    C. Organizations are pursuing the strategy of status quo.
    D. In today's context, organizations need leaders much more than managers in transforming them.
    E. None of these
34. Why did companies and universities develop programmes to prepare managers in such a large number?
    A. Companies and universities wanted to generate funds through these programmes.
    B. A large number of organizations were created and they needed managers in good number.
    C. Organisations did not want spend their scarce resources in training managers.
    D. Organisations wanted to create communication network through trained managers.
    E. None of these
35. Which of the following statements is NOT TRUE in the context of the passage?
    A. Bureaucratic culture can smother those who want to respond to changing conditions.
    B. Leadership produces change and has the potential to establish direction.
    C. Pressure on managers comes mostly from within.
    D. Leadership centres on carrying out important functions such as planning and problem-solving.
    E. Managers believe that they are the best and that their idiosyncratic traditions are superior.
36. Which of the following is not the characteristic of bureaucratic culture?
    A. Managers listen poorly and learn slowly.
    B. Managerial competencies are nurtured.
    C. Employees clearly see the forces that present threats and opportunities.
    D. Prevalence of unhealthy arrogance.
    E. Managers tend to stifle initiative and innovation.
37. Which of the following is SIMILAR in meaning to the word SMOTHER as used in the passage?
    A. Suppress B. encourage C. instigate
    D. Criticize E. attack
38. How has the author defined management?
    A. It is the process of adapting organizations to changing circumstances.
    B. It is the system of aligning people with the direction it has taken.
    C. It refers to creating a vision to help direct the change effort.
    D. Creating better performance
    E. Managerial training
39. Which of the following is SIMILAR in meaning of the word NURTURED as used in the passage?
    A. created B. developed C. thwarted
    D. surfaced E. halted
40. What, according to the author, is leadership?
    A. Process which keeps the system of people and technology running smoothly
    B. Planning the future and budgeting resources of the organization
    C. Inspiring people to realize the vision
    D. Carrying out the crucial functions of management
    E. None of these

Set C

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), Term II Examination, 2017-18

Subject: English for Professional Purposes–II — Subject Code: AHE 4083

Time: 45 Minutes — Max. Marks: 40

VERBAL APTITUDE

Choose the most appropriate option for Q. No. 1-6.

1. The word itself is so rare as to be almost obsolete. Which one is the adverb?
   A. theB. Itself C. so
   D. obsolete E. None of the above
2. She has a wheelchair that was _____ made for her.
   A. specially B. Especially C. most
   D. nothing E. none of the above
3. _____.the votes been counted _____they started breaking promises.
   A. Hardly do/ that B. Hardly did/ than
   C. Hardly had/ when D. Hardly did/when
   E. none of the above
4. The programmer's techniques were _____. The manager said that his work was_____ done.
   A. well/good B. good/well C. good/good
   D. well/well E. none of the above
5. An adverbial can modify a/an
   A. verb B. Adjective C. adverb
   D. clause E. all of the above
6. ____ those words ____ .all the newspapers published them.
   A. As soon as he had uttered/ that
   B. No sooner had he uttered/ than
   C. No sooner he had uttered/ than
   D. Although he uttered/ but
   E. none of the above

Decide which sentence is the most preferable in respect to grammar/ meaning and usage, suitable for formal writing.

7. A. The minister brushed away suggestions that he had considered resigning.
   B. The minister brushed off suggestions that he had considered resigning.
   C. The minister brushed in suggestions that he had considered resigning.
   D. The minister brushed on suggestions that he had considered resigning.
   E. The minister brushed of suggestions that he had considered resigning.
8. A. The allies were threatening to bring up the government by withdrawing from the ruling coalition.
   B. The allies were threatening to bring in the government by withdrawing from the ruling coalition.
   C. The allies were threatening to bring into the government by withdrawing from the ruling coalition.
   D. The allies were threatening to bring down the government by withdrawing from the ruling coalition.
   E. The allies were threatening to bring under the government by withdrawing from the ruling coalition.
9. A. Doctor advised Shreya to add pasta to the salads to bump upon her fibre intake.
   B. Doctor advised Shreya to add pasta to the salads to bump on her fibre intake.
   C. Doctor advised Shreya to add pasta to the salads to bump up her fibre intake.
   D. Doctor advised Shreya to add pasta to the salads to bump in her fibre intake.
   E. Doctor advised Shreya to add pasta to the salads to bump into her fibre intake.
10. A. Vimal cut down on coffee and cigarettes, and ate a balanced diet.
    B. Vimal cut in on coffee and cigarettes, and ate a balanced diet.
    C. Vimal cut up on coffee and cigarettes, and ate a balanced diet.
    D. Vimal cut under on coffee and cigarettes, and ate a balanced diet.
    E. Vimal cut on coffee and cigarettes, and ate a balanced diet.
11. A. I take away my mother. We are both impatient.
    B. I take on my mother. We are both impatient.
    C. I take in my mother. We are both impatient.
    D. I take after my mother. We are both impatient.
    E. I take upon my mother. We are both impatient.

Choose the correct option.

12. One who is easily deceived
    A. Eligible B. Negligible C. Gullible
    D. workable E. None of the above
13. The soldiers on foot.
    A. Infantry B. Infantile C. Infertile
    D. Indolent E. None of the above
14. A substance which kills germs
    A. Antidote B. Anti-Semitic C. Antigen
    D. Antiseptic E. None of the above
15. Absence of government.
    A. Archery B. Anarchy C. Monarchy
    D. Archaic E. None of the above
16. The dead body of an animal.
    A. Corpus B. Canvas C. Carcass
    D. Carnal E. None of the above

Pick out the most effective word(s) to make the sentence complete.

17. The Jurassic _____ the geological history _____for approximately 62 billion years.
    A. period since, lasts B. period over, last
    C. period in, lasted D. period for, lasting
    E. period at, have lasted
18. Every Monday, Sally ____ her kids to the football practice.
    A. Is driving B. Has been driving
    C. Drives D. Has driven E. Drove

Read each sentence to find out whether there is any grammatical or idiomatic error in it.

19. Advocates of non-violence (A)/ have always adviced (B)/ peaceful solutions (C)/ to the most complex problems (D)./ No error(E)
20. The farmer (A)/ felt badly about (B)/ losing all of his crop (C)/ overnight (D)./ No error (E)
21. Some highly significant (A)/ observations has been (B)/ made by the former UN secretary General, Mr. Perez De cueler, (C)/ in his annual report submitted to the current session of General assembly (D)./ No error (E)
22. One of the developing or underdeveloped (A)/ countries are not in favour of launching (B)/ chemical warfare because it trembles (C)/ to imagine its repercussions (D)./ No error (E)

Fill in the blanks with the most suitable option.

23. If you _____ the car, it _____ down in the middle of the forest.
    A. had checked/ would have broken
    B. had checked/ wouldn't have broken
    C. had been checked/ would have been broken
    D. had not been checked/ would not have been broken
    E. None of the options given is correct.
24. I _____ speak to him again unless he _____.
    A. won't/apologises B. will not/apologise
    C. would not/apologise
    D. would not have/had apologised E. None of these
25. I _____ out tonight if I _____ to work tomorrow.
    A. will go/didn't B. would go/ didn't

C. would go/ didn't have D. would go/ cannot
E. will go/cannot

26. If the trip _____ so expensive I _____ a reservation.
A. had/would have make
B. had been/would have been made
C. hadn't been/would have made
D. hadn't/ would have made
E. None of these

Choose the correct option.

27. I took that person_____ a thief.
A. of B. to C. about
D. for E. none of these

28. My first lesson ____ forgiveness came from my father.
A. in B. on C. upon
D. for E. none of these

29. I prefer this book ____ that.
A. to B. than C. then
D. for E. none of these

30. Rohan is ____ the campus.
A. in B. on C. upon
D. into E. none of these

31. The idiom "to be on the same page" means:
A. At the same place B. At the same time
C. To be in the same situation
D. To understand each other E. None of the above

32. The idiom "Back to Square one" means:
A. Back to the start B. To begin from the end
C. To go back to one's home D. To begin work again
E. None of the above

Read the passage carefully and answer the questions that follow.

In the second week of August 1998, just a few days after the incidents of bombing the US embassies in Nairobi and Dar-es-Salaam, a high-powered, brain-storming session was held near Washington D.C., to discuss various aspects of terrorism. The meeting was attended by ten of America's leading experts in various fields such as germ and chemical warfare, public health, disease control and also by the doctors and the law enforcing officers. Being asked to describe the horror of possible bio-attack, one of the experts narrated the following gloomy scenario. A culprit in a crowded business centre or in a busy shopping mall of a town empties a test tube containing some fluid, which in turn creates an unseen cloud of germ of a dreaded disease like anthrax capable of inflicting a horrible death within 5 days on any one who inhales it. At first 500, or so victims feel that they have mild influenza which may recede after a day or two. Then the symptoms return again and their lungs start filling with fluid. They rush to local hospitals for treatment, but the panic-stricken people may find that the medicare services run quickly out of drugs due to excessive demand. But no one would be able to realize that a terrorist attack has occurred. One cannot deny the possibility that the germ involved would be of contagious variety capable of causing an epidemic. The meeting concluded that such attacks, apart from causing immediate human tragedy, would have dire long-term effects on the political and social fabric of a country by way of ending people's trust on the competence of the government. The experts also said that the bombs used in Kenya and Tanzania were of the old-fashion variety and involved quantities of high explosives, but new terrorism will prove to be more deadly and probably more elusive than hijacking an aeroplane or a gelignite of previous decades. According to Bruce Hoffman, an American specialist on political violence, old terrorism generally had a specific manifesto-to overthrow a colonial power or the capitalist system and so on. These terrorists were not shy about planting a bomb or hijacking an aircraft and they set some limit to their brutality. Killing so many innocent people might turn their natural supporters off. Political terrorists want a lot of people watching but not a lot of people dead. —Old terrorism sought to change the world while the new sort is often practiced by those who believe that the world has gone beyond redemption!, he added. Hoffman says, —New terrorism has no longterm agenda but is ruthless in its short-term intentions. It is often just a cacophonous cry of protest or an outburst of religious intolerance or a protest against the West in general and the US in particular. Its perpetrators may be religious fanatics or diehard opponent of a government and see no reason to show restraint. They are simply intent on inflicting the maximum amount of pain on the victim.

33. In the context of the passage, the culprit's act of emptying a test tube containing some fluid can be classified as
A. a terrorist attack
B. an epidemic of a dreaded disease
C. a natural calamity
D. panic created by an imaginary event
E. None of these

34. In what way would the new terrorism be different from that of the earlier years?
(a) More dangerous and less baffling
(b) More hazardous for victims
(c) Less complicated for terrorists
A. (a) and (c) only B. (b) and (c) only
C. (a) and (b) only D. All the three
E. None of these

35. What was the immediate provocation for the meeting held in August 1998?
A. the insistence of America's leading
B. the horrors of possible bio-attacks
C. a culprit's heinous act of spreading germs
D. people's lack of trust in the government
E. None of these

36. What could be the probable consequence of bio-attacks, as mentioned in the passage?
(a) several deaths
(b) political turmoil
(c) social unrest
A. (a) only B. (b) only C. (c) only
D. (a) and (b) only E. None of these

37. The author's purpose of writing the above passage seems to explain
A. the methods of containing terrorism
B. the socio-political turmoil in African countries
C. the deadly strategies adopted by modern terrorists
D. reason for killing innocent people
E. the salient features of terrorism of yesteryear

38. According to the author of the passage, the root cause of terrorism is
(a) religious fanaticism
(b) socio-political changes in countries
(c) the enormous population growth
A. (a) only B. (b) only C. (c) only
D. (a) and (b) only E. All the three

39. The phrase —such attacks, as mentioned in the last sentence of the second paragraph, refers to
A. the onslaught of an epidemic as a natural
B. bio-attack on political people in the government
C. attack aimed at damaging the reputation of the government
D. bio-attack maneuvered by unscrupulous elements
E. None of these

40. The sole objective of the old terrorism, according to Hoffman, was to
A. plant bombs to kill innocent people
B. remove colonial power or capitalist system
C. make people realize the incompetence of the government
D. give a setback to socio-political order
E. None of these

Paper Code- AHE4081

SET-C

GLA University

Soft Skills (Even Sem)

Second Mid Term Examination (2017-18)

Course- B.Tech **Year- II**

Branch- All Branches

Instructions :

1. **There are 2 sections in the paper.**
2. **Attempt all questions. The maximum marks for the test are 20.**
3. **The total time allocated is 60 minutes.**
4. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
5. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
6. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
7. **Only black& blue pen should be used to fill forms.**
8. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
9. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

Signature of Invigilator-.....

S.N	Branch	Code
1	CS	1500
2	CE	1000
3	ECE	1300
4	EN	1400
5	EE	1100
6	ME	1200
7	MCA	8840

All The Best

Section-1 (Quantitative Aptitude)- No of Questions-20, marks-10

Q1. If the income tax be reduced from 3½% to 3⅓% what difference does it make to a man whose annual income in Rs 8400?
A] Rs 7 B] Rs 28 C] Rs 21 D] Rs 14

Q2. A man spent 12.50% of his money and after spending 75% of the remainder, he had Rs175 left. How much had he at first?
A] Rs800 B] Rs1200 C] Rs1600 D] None of these

Q3. Two numbers are respectively 20% and 50% more than a third number. What percentage is the first of the second?
A] 60% B] 70% C] 80% D] 40%

Q4. Candidate who gets 30% of the marks in an examination fails by 50 marks. Another candidate who gets 320 marks fails by 30 marks. Find the maximum number of marks.
A] 800 B] 900 C] 1000 D] None of these

Q5. In an examination paper is set to 2500 pupils of whom, one fifth are girls and the rest boys. 5% of the boys fail and 40% of the girls fail. What percent of the whole passed?
A] 82% B] 75% C] 80% D] 88%

Q6. A shopkeeper allows a discount of 12.5% on the marked price of a certain article and makes a profit of 20%. If the article cost the shopkeeper Rs. 210, what price must be marked on the article?
A] Rs. 280 B] Rs. 288 C] Rs. 300 D] None of these

Q7. A Camera shop allows a discount of 10% on the advertised price of a camera. What price must be marked on the camera, that costs him Rs. 600, so that he makes a profit of 20%?
A] Rs. 800 B] Rs. 720 C] Rs. 750 D] Rs. 850

Q8. In the land of the famous milkman Merghese Durian, a milkman sells his buffalo for Rs. 720 at some profit. Had he sold his buffalo at Rs. 510, the quantum of the loss incurred would have been double that of the profit earned. What is the cost price?
A] Rs. 600 B] 625 C] 675 D] None of these

Q9. By selling an article, Ram earned a profit equal to one-fifth of the price he bought it. If he sold it for Rs. 360, what was the cost price?
A] Rs281.75 B] Rs300 C] Rs312.50 D] Rs350

Q10. If SP of 10 articles is equal to cost price of 8 articles, find Profit or Loss Percentage
A] 25% Profit B] 25% loss C] 20% Profit D] 20% loss

Q11. If Rs.2000 amounts to Rs.2880 in 2 years at compound interest, what is the rate of interest per annum if the interest is being compounded annually?
A] 10% B] 20% C] 15% D] 25%

Q12. Find the interest earned in the first year on Rs.400 at 20%p.a. compound interest, the interest being compounded half yearly.
A] Rs.42 B] Rs.72 C] Rs.84 D] Rs.144

Q13. A sum doubles in 8 years at compound interest. In how many years will the sum become 4 times the original sum if the interest is compounded annually?
A] 16 B] 24 C] 64 D] 32

Q14. Which of the following rates of interest yield the maximum interest in 2 years on a certain sum?
A] Interest compounded per month at 1% p.m.
B] Interest compounded per quarter at 3% per quarter
C] Interest compounded per half year at 6% per half year
D] Interest compounded per year at 12% p.a.

Q15. A sum was lent at 20%p.a compound interest, the interest being compounded annually. Rs.1200 was paid back after 1 year. After another year Rs. 1440 was repaid to clear the loan. Find the sum lent.
A] Rs.8000 B] Rs.6000 C] Rs 2000 D] Rs 4000

Q16. What would a sum of Rs.8800 amount to in 16 years at a simple interest rate of 12% every year?
A] Rs.14440 B] Rs.18846 C] Rs.25696 D] Rs.32322

Q17. A sum of money invested at simple interest amounts to Rs 2480 at the end of four years and Rs.4080 at the end of eight years. Find the principal.
A] Rs.2040 B]Rs. 1480 C] Rs.1240 D] Rs.880

Q18. A, B and C play cricket. A's runs are to B's runs and B's runs are to C's as 3:2. They get altogether 342 runs. How many runs did A make?
A] 162 B] 108 C] 72 D] None of these

Q19. The ratio of the ages of A and B is 9:5. After 8 years B would reach the present age of A. Find the present ages of B and A respectively
A] 9, 5 B] 10, 18 C] 18, 10 D] 5, 9

Q20. The monthly salaries of two persons are in the ratio of 4:7. If each receives an increase of Rs.25 in the salary, the ratio is altered to 3: 5. Find their respective salaries.
A] 120 & 210 B] 80 & 140 C] 180 & 300 D] 200 & 350

Section-2 (Reasoning Ability)- No of Questions-20, marks-10

Q21. In the following alphabet second half is written first and then the first half is arranged in reverse order. Then which of the following will be the 5th letter to the left of 16th letter from the right?
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A.) S B.) T C.) P D.) F

Q22. How many pairs of letters are there in the word " **CASTRAPHONE**" which have as many letters between them in the word as in the alphabet?
A.) 3 B.) 4 C.) 5 D.) 6

Q23. From the word **'POSTSCRIPT'**, how many independent words can be made without changing the order of the letters and using each letters only once?
A.) 1 B.) 2 C.) 3 D.) 4

Directions (Q24 to Q25):
Instruction: Study the following arrangement carefully and answer the questions given below.
B 5 R 1 @ E K 4 F 7 © D A M 2 P 3 % 9 H 1 W 8 * 6 U J $ V Q #

Q24.Which of the following is the sixth to the left of the seventeenth from the left end of the above arrangement?

A.) © B.) 7 C.) D D.) A

Q25.Four of the following five are alike in a certain way based on their position in the above arrangement and so form a group. Which is the one that does not belong to that group?

A.) PM3 B.) KFE C.) 6J* D.) 7D4

Q26. 2, 10, 40, 120,____

A.) 240 B.) 360 C.) 470 D.) 210

Q27. 20 20 17 17 14 14 11 What numbers should come next ?

A.) 8, 8 B.) 8, 9 C.) 11 ,8 D.) 8, 9

Q28. 2, 10, 26, 50, ___, 122.

A.) 82 B.) 90 C.) 80 D.) 62

Q29. Look at this series: 14, 28, 20, 40, 32, 64, . . .
What number should come next ?

A.) 52 B.) 56 C.) 86 D.) 98

Q30. Find the missing number in the following series.
2, 3, 6, 15, 45, ?

A.) 95 B.) 105.5 C.) 147 D.) 157.5

Q31. Anil wants to go the university. He starts from his house which is in the East and comes to a crossing. The road to his left ends in a theatre, straight ahead ends in the hospital. In which direction is the University?

A.) East B.) North C.) South D.) West

Q32. Anitha started from home and walked some distance towards North-East and then turned 90 degrees left. After sometime she again turned 90 degrees left. In what direction is she walking now?

A.) North-West B.) South-West C.) South D.) East

Q33. In a certain code LEARNING is written as LGNINRAE. How will SURPRISE be written in that code?

A.) ESRIPRUS B.) RUSEPSIR C.) SESIRPRU D.) ESIRPRSU

Q34. If PRIVATE is coded as 1234567 and RISK is coded as 2398, how is RIVETS coded?

A.) 232679 B.) 243769 C.) 234769 D.) 234976

Q35. Pointing to Sahil, Neeru says, "I am the daughter of the only son of his grandfather." How Neeru is related to Sahil?

A. Daughter B. Mother C. Sister D. Aunt

Directions (36 to 37):
Each of these questions is based on the following information:
M % N means M is the son of N.
M @ N means M is the sister of N.
M $ N means M is the father of N.

Q36. Which of the following shows the relation that C is the granddaughter of E?

A.C % B $ F $ E B.B $ F $ E % C C.C @ B % F % E D.E % B $ F $ C

Q37. Which of the following shows the relation that S is the father of Q?
A.S @ P $ Q B.Q @ P % S C.Q $ S @ P D.None of these

Q38. In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row?
A. 8 B. 9 C. 10 D. 11

Q39. In a certain code, '37' means 'which class' and '583' means 'caste and class'. What is the code for 'caste' ?
A.7 B.3 C. Either 5 or 3 D. Either 5 or 8

Q40. Ramesh walks 30 m towards south. Then turning to his right he walks 30m. Then turning to his left he walks 20m. Again turning to his left he walks 30m. How for is he from his starting position?
A.) 30m B.) 20m C.) 50m D.) None of these

Data Anal

Mid Term Examination, 2017-18
Operating Systems (CSE4001)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What is the difference between direct and indirect communication?

II. What is race condition?

III. What are conditions under which a deadlock situation may arise?

IV. Define is the role of claim edge in resource allocation graph algorithm.

V. What is the difference between binary semaphore and counting semaphore?

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Discuss the concept of hardware synchronization using Test and Set instructions in Multiprocessor environment.

II. Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned.

Method Used by P1	Method Used by P2
while (S1 == S2) ;	while (S1 != S2) ;
Critical Section	Critical Section
S1 = S2;	S2 = not (S1);

State that the above code segments must satisfy the mutual exclusion and progress conditions for critical section problem or not.

III. A system has two processes and three identical resources. Each process needs a maximum of two resources. Is deadlock possible? Explain your answer.

IV. Explain how the system can recover from the deadlock using (a) Recovery through rollback, and (b) Recovery through killing processes.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Critically evaluate the method of message passing as a mean of inter-process communication. Also discuss implementation issues in message passing system for inter-process communication.

II. Consider the following snapshot of the system:

	Allocation	Max	Available
	A B C D	A B C D	A B C D
P1	0 0 1 2	0 0 1 2	1 5 2 0
P2	1 0 0 0	1 7 5 0	
P3	1 3 5 4	2 3 5 6	
P4	0 6 3 2	0 6 5 2	
P5	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

a. Is the system being in Safe state or not?

b. If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately.

III. Explain Reader Writers problem with its semaphore solution and specify how it meets all three requirements of solution to its critical section problem.

IV. If a system has only two processes, does it make sense to use a barrier to synchronize them? If yes, explain your answer with solution for synchronize them.

B.Tech II year C.S. Even CS
II-mid Tester

Mid Term Examination, 2017-18
Course: B.Tech Year- II: Semester: IV
Discrete Mathematics (CSE 4005)

Uni. Roll No:

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5 marks)**

1. When an inference system is said to be sound and complete? Explain.
2. Is it possible to distribute quantifiers in predicate logic? Justify your answer.
3. If P and Q are two propositions. Which of the following logical expressions are equivalent?

   1. $P \vee \sim Q$
   2. $\sim(\sim P \wedge Q)$
   3. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$
   4. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$

4. Express the following English sentence in predicate logic. Make the domain of your variables to be the set of all objects.
   "Two people are siblings if they have the same mother or the same father".
5. Consider a company that assembles computers. The probability of a faulty assembly of any computer is p. The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q. What is the probability of a computer being declared faulty?

Section B

Note: Attempt any three questions **(2x3=6 marks)**

1. What are the application areas of Bayes Theorem in computer science. Researchers surveyed 100 students on which superpower they would most like to have. This two way table displays data for the sample of students who responded to the survey:

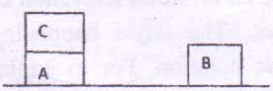
Superpower	Male	Female	Total
Fly	26	12	38
Invisibility	12	32	44
Other	10	8	18
Total	48	52	100

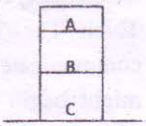
A Student will be chosen at random. find the probability that the student was male, given the student chose to fly as their superpower.

2. i)Use the logical equivalence laws to show that $\neg(P \vee \neg(P \wedge Q))$ is a contradiction.
ii)Test the validity of the argument:
"If my brother stands first in the class, I give him a watch. Either be stood first or I was out of station. I did not give brother a watch this time. Therefore, I was out of station".

3. Using the inference rules show that S can be inferred if the following logical sentences are believed.
1.P
2.$Z \vee H$
3.$H \vee \neg P$
4.$S \vee \neg(\neg K)$
5.$H \rightarrow K$
6.$K \rightarrow (S \vee \neg P)$
7.$(Z \wedge \neg H) \rightarrow K$
8.$(K \wedge \neg Q) \rightarrow S$

4. Consider three blocks A, B and C which can be stacked on top of each other, or on a table, for instance, in configurations like the following:





In this problem, we do not move the blocks around, but rather consider a fixed configuration of the blocks. In such a configuration, let $S_{A,B}$ be a proposition symbol representing the fact that A is stacked above B, and similarly define $S_{B,A}$, $S_{A,C}$, $S_{C,A}$, $S_{B,C}$ and $S_{C,B}$. For instance, in the figure on the left, $S_{C,A}$ would be the only one of these symbols that is true. In the figure on the right, the only true symbols are $S_{A,B}$, $S_{B,C}$ and $S_{A,C}$. For parts (a-b) below, be sure to explicitly list all of the required clauses or sentences. Do not just give examples of a general pattern.

a) Show how to represent using propositional logic the fact that if one block is stacked above another, then the second cannot be above the first (for instance, if A is above B, then B cannot be above A).

b) Show how to represent the fact that if one block is stacked above a second block, which is in turn stacked above a third block, then the first block must be stacked above the third block (for instance, if A is above B, and B is above C, then A must be above C).

Section C

Note: Attempt any three questions **(3x3=9 marks)**

1. A company buys 70% of its computers from company X and 30% from company Y. Company X produces 1 faulty computer per 5 computers and company Y produces 1 faulty computer per 20 computers. A computer is found faulty what is the probability that it was bought from company X?
2. For every formula Ø of prepositional logic, there exist an equivalent formula Ψ in conjunctive normal form (i.e., a formula Ψ such that Ø= Ψ).This conversion leads us for an easy test of validity of formula. What you can say about the time complexity in worst case to show that a formula is valid. Also convert the following expression into CNF form.

$$\neg((\neg P \rightarrow \neg q) \wedge \neg r)$$

3. A media tycoon has an ideaa for an all-news television network called TNN: The Taza News Network. The day's happenings are to be communicated concisely in logic notation. For example, a broadcast might begin as follows:"THIS IS TNN. The domain of discourse is {Ram, Badri, Laxmi, Drupad, Triloki}. Let D(x) be a predicate that is true if x is deceitful. Let L(x,y) be a predicate that is true if x likes y. Let G(x,y) be a predicate that is true if x gave gifts to y."
Translate the following broadcasted logic notation into English statements:

i. $(\neg(D(Badri) \vee D(Drupad))) \rightarrow (L(Ram, Badri) \wedge L(Badri, Ram))$

ii. $\forall x(x=Laxmi \wedge \neg L(x, Triloki)) \vee (x \neq Laxmi \wedge L(x, Triloki)) \wedge \forall x(x=Drupad \wedge L(x, Laxmi)) \vee (x \neq Drupad \wedge \neg L(x, Laxmi))$

iii. $\neg D(Laxmi) \rightarrow (G(Ram, Badri) \wedge \exists x\, G(Badri, x))$

4. In the Computer Science (CS) Department of GLA University, 200 students registered for the course CSE 4004 (Object Oriented Programming). 30% of the registered students are CS majors and the rest are non-majors. From the student registration data we observe that 80% of the CS majors are males, where as only 40% of non-majors are males. Find the following:

1.The probability that a randomly selected student is a CS major.

2.The probability that the selected student is a CS major and a male.

3.The probability that the selected student is a male.

MID TERM EXAMINATION 2017-18

Computer Networks (CSE 4003)

Course: B.Tech **Year: II** **Semester: IV**

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Note: Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.

Section-A

Note: Attempt all Questions **(1X5=5)**

Q1. An IPv4 packet has the first few hexadecimal digits as shown below

0X4500005C000300005906....

How many hops can this packet take before being dropped?

Q2. An organization has a class-B network and wishes to form subnets for 24 departments. What will be the subnet mask?

Q3. Explain the purpose of FCS field in PPP frames.

Q4. Bit-stuff the given data

10011111110011111010011111111110011111

Q5. An 8-bit byte with binary value 10101111 is to be encoded using an odd-parity Hamming code. What is the binary value after encoding?

Section-B

Note: Attempt any three questions **(2x3=6)**

Q1. A Sliding window protocol of 4Mbps point to point link has propagation delay of 0.5sec. Assume that each frame carries 2KB of data. What is the minimum no. of bits used for sequence number field?

Q2. A sender needs to send the four data items Ox3456, OxABCC, Ox02BC, and OxEEEE. Answer the following:

a. Find the checksum at the sender site if there is no error.

b. Find the checksum at the receiver site if the second data item is changed to OxABCE.

Q3. Suppose that the timeout values for Go-Back N and SR protocols are sufficiently long such that 7 consecutive data segments and their corresponding ACKs can be received (if not lost in the channel) by the receiving host (Host B) and the sending host (Host A) respectively. Suppose Host A sends 7 data segments to Host B, and the 4th segment (sent from A) is lost. In the end, all 7 data segments have been correctly received by Host B. How many segments has Host A sent in total and how many ACKs has Host B sent in total? What are their sequence numbers?

Q4. Consider a binary code that consist of only four valid codewords as given as 00000,01011,10101,11110. Let the minimum Hamming distance of the code be x and the maximum number of erroneous bits that can be corrected by the code be y. Determine the values of x and y.

Section-C

Note: Attempt any three questions **(3x3=9)**

Q1. Explain the two node instability problem in Distance Vector Routing. Also explain the solution to this problem.

Q2. An Internet Service Provider (ISP) is granted a block of addresses starting with 145.75.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows:
(a) The first group has 128 customers; each needs 256 addresses.
(b) The second group has 128 customers; each needs 64 addresses.
(c) The third group has 64 customers; each needs 128 addresses.
Design the sub blocks and find out how many addresses are still available after these allocations.

Q3. Explain the following fields in IP Header clearly mentioning the size of each field
(a)TTL (b) Protocol (c) Flags

Q4. Explain the four types of links defined by OSPF? Why do OSPF messages propagate faster than RIP messages?

Mid Term Examination, 2017-18
Introduction to Microprocessors (CSE4002)

Time: 1.5 Hrs. **Total Marks: 20**

Uni. Roll No: .....

Section-A

Note: Attempt all Questions (1*5=5)

I. What is the use of Direction flag in 8086 µp?

II. Explain the mathematical function that is performed by the following instructions:

```
MVI  A, 07H
RLC
MOV  B, A
RLC
RLC
ADD  B
```

III. Write any two signals with their functionality used in maximum mode operation in 8086 µp.

IV. Which instruction that changes arithmetic sign of a signed nun from positive to negative or from negative to positive.

V. If value of base register is 4FF0H and value of offset is 00F4H. What will be physical address generated in 8086 µp?

Section-B

Note: Attempt any three questions (2*3=6)

I. Write assembly language program for generating delay of 2 ms and Calculate the value of counter (in Hexadecimal) for delay 2ms by using 8085 microprocessor, assuming the system clock period is 0.33 µs.

II. How 8086 microprocessor support parallel processing? Explain the concept with suitable diagram.

III. Calculate the delay (in ms) in the following loop, assuming the system clock period is 0.5 μs:

Label	Mnemonics	8085 T-States
	MVI D, 10H	4
NEXT:	LXI B, 1000H	10
DELAY:	DCX B	6
	MOV A, C	4
	ORA B	4
	JNZ DELAY	10/7
	DCR D	4
	JNZ NEXT	10/7

IV. How stack and subroutine support in assembly language programming? Discuss CALL and RET instruction with example.

Section-C

Note: Attempt any three questions (3*3=9)

I. Specify the number of times the following loops are executed. Also calculate net time delay with frequency of processor 5MHz.

i)
```
    MVI  A, 17H   (4 T-States)
L1: ORA  A        (4)
    RAL           (4)
    JNC  L1       (10/7)
```

ii)
```
    MVI  B, 64H   (4)
L2: NOP           (4)
    DCR  B        (4)
    JNZ  L2       (10/7)
```

iii)
```
    LXI  B, 1000H (10)
LOOP: DCX B       (6)
    JNZ  LOOP     (10/7)
```

II. Explain functional components of 8086 microprocessor with suitable diagram.

III. Write assembly language program to count, how many values in a given list (size 20 in decimal) are greater than the value stored in register B. Starting address of the list is 2000H.

IV. Explain addressing modes of 8086 μp with example.

Mid Term Examination, 2017-18
Object Oriented Programming (CSE4104)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Enlist the different scenarios causing "Exception in thread main"?

II. Differentiate between Checked and Unchecked exceptions in Java? Name the base class of Exception class.

III. What is deadlock in multithreading? What measures you should take to avoid deadlock?

IV. What is daemon Thread? Name the method used to check whether a Thread is Daemon or not?

V. Write complete syntax for loading and displaying image on an applet? Name any four methods of Graphics class with arguments description?

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain different ways of creating a thread with complete syntax declaration for each. Which one would you prefer and why?

II. Explain the life cycle of Applet? Differentiate between Java Applet & Java Application?

III. Discuss the significance of the main Thread in Java? Design a program to assign a name and priority to a thread and display the assigned name and priority.

IV. Explain thread state diagram? Can we call the run() method instead of start()? Justify your answer with an example?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Design a GUI in Applet/Frame with following five components and given constraints:
Components: Label, TextField, Checkbox, RadioButton and Button

Constraints: Background color : blue, Font: Tahoma. size=20,style=Italic.

II. Explain Event Delegation Model describing appropriate components (Class and its Methods, Interface and its methods) for Button and Mouse events? Design a program to demonstrate Button click event on Applet /Frame.

III. Discuss the use of Layout Manager in java? Name the different Layout Managers used in java? What is the default layout of an Applet? Design a program to change default layout of an applet to some other layout.

IV. Differentiate between Button and JButton? Why swing components are called as lightweight component?

C3

Mid-Term Examination
Even Semester 2017-18

Course: B. Tech. IIYr. **Univ. No.....**
Title of paper :-Environmental studies **Paper code: AHS-3001** **Time: 1.5 hr**

Section –A

Note: Attempt All Questions **1x5 =5 marks**

Q1. Define Eutrophication.

Q2. What is Red data book?

Q3. Write the full form of SPM & PAN in context of pollution.

Q4. Define Pulverization.

Q5. Name the unit used for measuring Depletion in Ozone layer

Section –B

Note: Attempt Any Three Question **2x3 =6 marks**

Q1. Explain Malthus theory of population Growth.

Q2. Discuss both positive and negative impact of urbanization.

Q3. What are the ill-effects of modern agriculture Practices on environment?

Q4. Global warming is one of the serious environmental problems of concern. Commenting on this statement suggest some effective measures to reduce the impact of this problem.

Section –C

Note: Attempt Any Three Question **3x3 =9 marks**

Q1. What is biomass energy? Discuss its production and uses.

Q2. What are the ill-effects of solid waste? Discuss some prominent methods for the disposal of solid waste.

Q3. What is biodiversity Depletion? Suggest some effective conservation methods to reduce biodiversity depletion.

Q4. Define Noise Pollution. Explain its effects and control measures.